

Assignment 1

SoE, FoSTA

Department of Electronics and Communication Engineering

B. Tech. – ECE & VDT

Course Code: ECE/VDT 2108		Course: Computer Architecture & Processor	Semester: III
Date of Submission: By 04:00 PM on 18/08/2026 extended to 11 AM on 24/08/26			Max. Marks: 200
Instructions: All questions are compulsory. Missing data, if any, may be assumed suitably. Marks are indicative of word limit. <i>One mark = 20 words. Do not cross the word limit.</i>			
Q 1	Define the following terms:		2 x 6
	i. Digital Computer	iv. Microarchitecture	= 12
	ii. Computer Architecture	v. Microoperations	
	iii. Computer Organization	vi. Register Transfer Language	
Q 2	What distinguishes computer organization from computer architecture?		4
Q 3	Briefly explain about the different generations of computers. Tabulate & highlight the significant technological contributions & human contributors of each generation.		10
Q 4	List the various elements that constitute a computer? Explain the roles and responsibility of each element.		10
Q 5	In how many ways are computers divided? Distinguish between each type of computers.		10
Q 6	Describe the features of the Von Neumann architecture. Compare & contrast Von Neumann and Harvard architecture.		10
Q 7	Explain the working of the following. Support your answer with circuits formed with basic gates:		4 x 6
	i. D Flip-Flop	iv. Quadruple 2x1 multiplexer	= 24
	ii. 3-to-8 line Decoder	v. 4-bit register with parallel load	
	iii. 4-to-1-line multiplexer	vi. 4-bit Bidirectional Shift Register with Parallel Load	
Q 8	How are the +64 and -64 signed numbers represented in the signed-magnitude, 1's complement, and 2's complement number systems?		4
Q 9	Perform the addition of two signed numbers R1= +64 and R2= +84 using 2's complement number systems. The size of each register is 8-bit.		4
Q 10	Categorize the following statements as register or memory transfer statement. Explain the operation being performed.		4
	i) $A \leftarrow B$ ii) $A(L) \leftarrow B(H)$	iii) $R1 \leftarrow M[AR]$ iv) $M \leftarrow R2$	8
Q 11	Represent the following conditional control statement as a register transfer statement. If (S = 1) then (AL $\leftarrow$ BH). Explain the operation being performed.		4
Q 12	Represent the following conditional control statement as a register transfer statement. If (y = 1 && T2=1) then (R2 $\leftarrow$ R1) and (R1 $\leftarrow$ R2).		8

	Create a block diagram of the hardware that implements the register transfer statement. Explain the working of the circuit.													
Q 13	What specifically is a bus in the context of computer architecture. What is its role. List the names of five most popular such structures.	8												
Q 14	Draw a neatly labelled diagram of a common bus system, as per the constraints mentioned below, that facilitates the transfer of data between four registers each of which is 4-bits. Explain the working in each case. i) Multiplexers, ii) Tri-state buffers and a decoder.	8 x 2 = 16												
Q 15	What is micro-operation? What are the different types of micro-operations?	4												
Q 16	Design a 4-bit adder-subtractor combinational circuit using four full-adder circuits. Explain its working using the following combination. <table border="1"><tr><td>Case</td><td>M</td><td>A</td><td>B</td></tr><tr><td>i)</td><td>0</td><td>0111</td><td>0110</td></tr><tr><td>ii)</td><td>1</td><td>0101</td><td>1010</td></tr></table>	Case	M	A	B	i)	0	0111	0110	ii)	1	0101	1010	8
Case	M	A	B											
i)	0	0111	0110											
ii)	1	0101	1010											
Q 17	Design a 4-bit arithmetic-logic-shift circuit. Explain the working of the circuit for the following functions. <table><tr><td>i) Transfer A</td><td>iii) XOR</td></tr><tr><td>ii) Add with carry</td><td>iv) Shift right A</td></tr></table>	i) Transfer A	iii) XOR	ii) Add with carry	iv) Shift right A	8 + 8 = 16								
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ii) Add with carry	iv) Shift right A													
Q 18	The initial values of the 8-bit registers R1, R2, R3, and R4 are as follows. Determine the results in each register once the following sequence of micro-operations has been performed. i) $R1 \leftarrow R1 + R2$ ii) $R3 \leftarrow R3 \wedge R4, R2 \leftarrow R2 + 1$ iii) $R1 \leftarrow R1 - R3$ Given: $R1 = 11110010, R2 = 11111111, R3 = 10111001$, and $R4 = 11101010$.	8												
Q 19	Find the sequence of binary values in register R following a logical shift-right, a circular shift right, a logical shift left, and a circular shift left, starting from $R=11111111$.	8												
Q 20	Write short note on the following: i) RISC architecture and ii) CISC architecture.	10 x 2 = 20												